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Parametric Modeling of a Lunar Base Architecture with a Focus on Power Systems Approaches

Sonal Mehta*, Bhargavi V. Thakar* †, Bradford E. Robertson‡ and Dimitri N. Mavris§
Aerospace Systems Design Laboratory, Georgia Institute of Technology, Atlanta, GA, 30308

One of the goals of the Artemis program is to establish a lunar base on the lunar South Pole. A parametric model grounded in system dynamics has been developed for conducting time-dependent investigations of parameters. The objective is to gain insights into the dynamic functioning of the lunar base. The methodology involves performing a physical and functional decomposition. This is followed by a parametric breakdown of these systems which aids in identifying key input and output parameters along with their interactions. The parameters are then used to develop the tool. The capabilities of the tool are demonstrated through a sample problem which considers key mission components such as crew size, mission duration with lunar days and nights. The focus of the problem is the comparison between singular and multiple power generation sites. Results focus on mass comparison between the two trades. Conclusions are thus drawn about the more favorable scenario and future additions to enhance the functionality of the framework.

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I. Introduction

Lunar exploration has been on the forefront of space exploration in recent years. NASA's Artemis program aims to establish a sustained presence on the lunar surface [1–3]. New science research can be conducted on the moon to investigate the potential of utilizing local moon resources for energy generation and sustaining life [4–6]. The commercial space sector has also seen substantial growth in the recent years due to the surge in demand for lunar technologies.

The prospective location of the base camp will be on the lunar south pole due to ample sunlight throughout the year and relative ease of communication with Earth [[2],[7]]. Other aspects of the lunar base such as surface habitat design, its constituent systems and technologies, and the science activities that will be carried out remain unknown. This leaves several open questions about the lunar base architecture. A comprehensive study needs to be carried out to understand trade-offs of various architectures.

A lunar base is comprised of several highly interdependent systems and subsystems. The interactions between these systems and subsystems are non-linear and can be mapped as resource flow rates and feedback loops. To ensure the minimum viable operating conditions of the base, stocks of life-support products, such as water and oxygen, and their corresponding effects on other systems such as the power system must be tracked. Further, these systems and subsystems exhibit time-dependent behavior due to the varying operating conditions. An effective approach to incorporate these time-dependent features is through the utilization of a system dynamics model[8]. In the hostile lunar environment, it is crucial to ensure that systems meet operational requirements during the lunar night when sustenance becomes even more difficult, and variations over lunar day and night need to be accounted for. This highlights the necessity of a capability to understand day and night cycles at the base to aid the dynamic study.

Illumination maps contain data about the light and shadow conditions (which corresponds to lunar day and night) of the area within 2 degrees of the lunar South Pole [9]. The day-night data from illumination maps can be used to ensure that the operational criteria are being met over the day-night cycles. This paper details the development of a parametric tool designed for conducting trade studies on lunar base architecture, providing valuable insights on its implementation. The illumination map is employed to facilitate monitoring of systems, and its application is further elucidated through a sample problem, showcasing the tool's usage and capabilities.

*Co-First Author: Graduate Research Associate, Aerospace Systems Design Lab, Department of Aerospace Engineering
†*Co-First Author: Graduate Research Associate, Aerospace Systems Design Lab, Department of Aerospace Engineering
‡Research Engineer II, Aerospace Systems Design Lab, Department of Aerospace Engineering
§S.P. Langley Distinguished Regents Professor, Aerospace Systems Design Laboratory, School of Aerospace Engineering. Fellow AIAA.

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II. Modeling

The development of the tool followed depended on the initial functional and physical decomposition of the base requirements. A use case for the lunar base was used as a starting point for guiding the direction of tool development. The conops of this mission involves crew stay on the lunar base for ≃30 days, wherein the crew carries out basic scientific activities on the surface of the moon and within the habitat.

The baseline mission was deconstructed into essential functions, which drove the physical architecture of the lunar base. The parameterization of the physical system accounts for the inputs and outputs. Subsequently, the framework was tailored to these derived parameters.

A. Functional and Physical Breakdown Identification

The baseline mission helps identify the functions of the lunar base. The functional breakdown of the lunar base is illustrated in Fig. 1 on the left. These functions are carried out by multiple subsystems that can be grouped into larger systems. This encompasses the physical breakdown of the lunar base which can be seen on the right of Fig. 1. Of the systems shown, the modeling focuses on the power system, thermal system, environmental control and life support system (ECLSS), and in-situ resource utilization (ISRU) system.

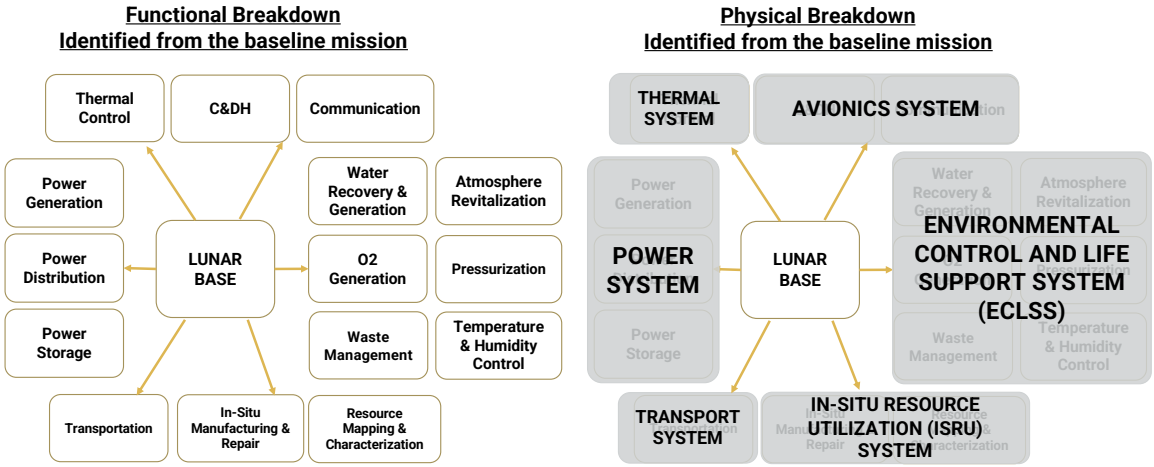


Fig. 1 Defining base Architecture in terms of Physical and Functional Breakdown

B. Parameterization of Physical Systems

Parameterization refers to the process of breaking down the various systems into their constituent parameters that describe the systems. The physical architecture shown in Fig. 1 was parameterized for the modeling process. The global parameters required by each system are the crew size, and mission duration. Each of the models have their own unique set of parameters that either define the functioning of their subsystems or are required to connect their constituent subsystems

The ECLSS and ISRU systems require inputs of production rates and frequencies, consumption rates and frequencies, system mass, cyclic or continuous operation modes, and power requirements for different modes of operation. The thermal system requires the heat load of all the lunar base systems. Additionally, technology specific parameters such as operating temperature of the radiators, the emissivity and absorptivity of radiators, number of radiator panels desired, heat pump head pressure, heat pump working fluid density, pump efficiency, and fluid flow rate are required. There are assumed values of power and mass for avionics systems based on the number of communication nodes. These values for the system don't change over time and thus do not contribute to the dynamic behavior of the base. The power system models the generation, distribution and storage of energy for the lunar base operation. It takes the power requirements of the other systems as an input and models the rest of the power system based on that one input. The illumination map also provides it with illumination data, which serves as an input for certain calculations. This is described in detail in

the modeling section.

After the parameters were identified, their interactions were mapped as shown in the N2 diagram in Fig. 2. The N2 diagram is not an indication of feed-forward and feedback loops, but is a pictorial representation of system connectivity for model development. It helps in accounting for the different resource connections and stocks. The model was then synthesized from the parametric breakdown and interactions and a system dynamics formulation was applied.

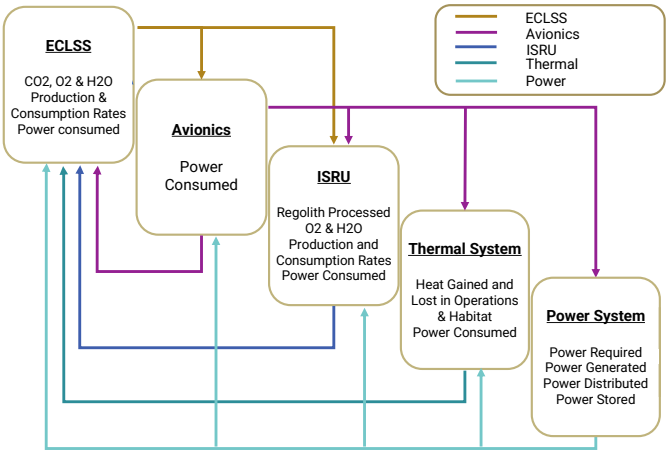


Fig. 2 Interactions between the various systems of the lunar base

C. System Modeling

Following the identification of the systems and their parameters that need to modeled, the tool is developed in Python. Each of the subsystem models accumulates the values of products or "stocks". The outputs provide a quantitative understanding of these cyclical resource dependencies. Some systems and subsystems have governing equations, while some have parameter relationships that can be varied within a specific range found in literature. These values are technology dependent and so each case has a predetermined set of technologies that are considered. Below are the descriptions of each of the models thus produced and the equations that govern their behavior.

1. ECLSS System Models

Environmental Control and Life Support System generates life support products and ensures that the environment within the habitat is suitable and comfortable for humans to live in. Each of the models within this system track the mass of the equipment required for functioning, the amount of product consumed during mission operation, amount of product produced during operation, and the power consumed during the desired operation settings. A feature across all subsystems within ECLSS is the option to determine cyclic or continuous operations for the equipment. The power consumption changes based on these operating modes, and these modes are inputs to the model. The cyclic operation mode is broken down into time of active operation and time in standby mode. The details of how inputs and formulae are applied to each subsystem are given below.

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CO2 Removal Assembly (CDRA)

This subsystem filters out the carbon dioxide in the air inside the habitat by absorbing it into a desiccant bed. That carbon-dioxide is later used in electrolysis to produce water and is thus linked to water production rates.

The CO_2 Removal Model first models the amount of carbon dioxide produced by the crew over the time. This takes the value of 0.9936 kg/CM-day (basal metabolic load) ([10]) as the CO_2 emitted by a human, and multiplies it by the number of crew. The next step involves modeling the amount of carbon dioxide removed by the assembly based on the partial pressure of CO_2, which is given by:

$$R = (0.158 \times PP_{CO_2} - 0.035) \times 0.453592 \times 24 \tag{1}$$

where R is the amount of carbon dioxide (in kg) removed in a day and PP_{CO_2} is the partial pressure of CO_2. This pressure is in mm of Hg, so a correction factor of 0.035 is applied for SI units.

The CO_2 levels are monitored by subtracting the amount removed from the amount produced daily, indicating how effectively the CDRA works. The amount of power consumed by the CDRA to remove the desired amount of CO_2 is found from the cycle time that the machine is set to run. For a higher crew-size or less EVAs in the mission, the CDRA would have to run longer to maintain CO_2 levels. It can also be set to "continuous operation" in which case, the power increase is demonstrated by a higher power required to effectively maintain the desired partial pressure of carbon dioxide within the habitat. The mass of the assembly is calculated based on an average of system masses reported in [11].

Water Recovery System (WRS)

This system model is derived from the WRS on the ISS where wastewater from human activities and urine are processed and filtered multiple times to be reused as potable water for drinking and other purposes [12]. This models the amount of water produced by the urine processing system and adds the amount of water that is generated from the CDRA's Sabatier process. This is done by using the appropriate molar mass conversions. The amount of urine processed is given by ([13, 14]):

$$\text{urine processed} = \frac{(8296/2) \times 0.96 \times 453.592}{(365.25 \times 24 \times 1000)} \tag{2}$$

where 0.96 is the expected efficiency of the urine processing bed. The denominator is for converting from g/year to kg/day. The formula is derived from the projected amount of urine that can be processed in the machine's operating lifetime [13].

The consumption is modeled as the nominal potable water consumption rate given in [10]. The amount consumed is 5.37984 kg/day nominally and is multiplied by the crew size. Ultimately, the levels of H_2O are tracked by adding the amount produced from urine and CDRA which is reduced by the amount consumed by crew. The power consumption depends on production frequency.

Oxygen Generation Assembly (OGA)

The oxygen generation assembly (OGA) for the lunar base is modeled as per the existing OGA on the ISS [15]. This oxygen is generated by electrolyzing the water produced by the WRS or from water stores. Generally, this process uses water from other processes such as CDRA and WRS to maintain a "closed-loop" type dependence within the ECLS System. It can also take water that is produced from ISRU Regolith Water-Ice Mining and recovery. The parametric model can accept either sources as inputs.

The oxygen production rate of the OGA is 9.2 kg/day, assuming peak production at continuous operation [15]. This amount is accumulated over the period of the mission, and is aggregated with the oxygen produced by ISRU. The oxygen levels can thus be monitored to see the if the levels are acceptable for human survival with redundancies. The power consumption is monitored based on the type of operation cycles.

2. ISRU System Models

In-Situ Resource Utilization contains models for functions related to the production of life-support products, oxygen and water, by processing lunar regolith. Regolith by itself has wide manufacturing uses, but this model, and study, is focused on monitoring the life-support products from regolith for minimum viable operations. There is a high-level accounting of the regolith excavation process and its direct usage for production [16]. All the technologies modeled in this subsection are in trial/experimental phases and their Technology Readiness Levels (TRLs) range from 3-6 [16, 17].

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Table 1 Oxygen Generation Processes in ISRU

Process	Load (kg/hr)	% Yield (by mass)	Power Consumed (kW)	System Mass (kg)
Carbothermal Reduction	15	15	13.5	450
Ilmenite Reduction	36.5	1.5	15	800
Electrolysis	12.5	20	66	550

A unique aspect of this model is the ability to factor in different production and usage rates for lunar day and lunar night (kg/day). This feature is unique to this system because all other functions are required at all times. ISRU, however, still has experimental processes and depends on processes that have variable outputs from lunar day to lunar night based on the conditions on the lunar surface, external to the lunar habitat. This flexibility allows for more case studies and experiments to be run to understand the impact of ISRU and understand its dependencies. Along with resource rates, it also takes as an input the frequencies of product production (1/no. of days) and consumption which accounts for processes that are not run everyday, especially in the initial stages of ISRU implementation.

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Regolith Excavation

Regolith is the main resource on the moon being studied. There are several potential uses for it from producing oxygen, to manufacturing materials that can be used for construction of parts of the lunar habitat. The excavation process is accounted for in the model since production rates for other processes depend on it. The regolith excavation rate is kept constant through the duration of the day and a separate constant for the night. The frequency of excavation determines the accumulation of product throughout the operation period.

Nominal value for regolith extraction is 2700 kg/day for lunar day as indicated by [18]. This assumes that the rover/excavator would be able to extract that much at a midway technology maturation point. The amount of consumedregolith is assumed to be a factor of the produced regolith. It is assumed that the amount of regolith consumed is lesser than that excavated per day, so the stock of regolith increases linearly.

The amount of power consumed would change depending on whether the equipment is run during lunar day or lunar night, but there is no regression data available to model these values.[16, 17, 19]. So, preliminary data is used, which may or may not be accurate for a fully sized system.

Water Extraction from Regolith

Water can be extracted from the lunar soil in the permanently shaded regions (PSR) from the water ice that has been detected there [20]. Thermal mining or the process of regolith extraction with certain catalysts can help extract the water from this. A differing amount of water can be retrieved from the soil depending on the composition of soil that was excavated. This differing amount is modeled as a percentage of the regolith that is extracted, regolith mass being 2700 kg/day/rover [18]. The Sabatier process can be used for production as well which would result in a different amount of water. Just like the other subsystems, this computes the power consumed by the water processing machines and calculates the mass of the entire system.

Oxygen Generation from Regolith

There are multiple processes that are proposed for oxygen generation using regolith. These are all derived from the water that is extracted from the regolith, so the production amount is directly related to the amount of water that can be extracted [21]. However, there are more technology demonstrations for O_2 generation, and so regression values are used for this process. Each of the processes have varying efficiencies and yields with differing power consumption and system masses [22]. A single model is developed which accumulates the amount of O_2 over time with a fixed rate of production per day (with separate values for lunar day and lunar night). The loads, resulting yields, power consumed and system masses are listed in the table 1 and have been taken from [23, 24]

3. Avionics System Model

The avionics system for lunar base is comprised of three subsystems, namely, Communications, Command & Data Handling (C&DH) and Guidance Navigation & Control (GNC). The communication system is the base's connection to Earth, and allows data and information to be shared with mission control. The C&DH controls the connections between

systems and within systems, and tracks all data for those [25, 26]. These systems are capable of processing, storing, and managing vast amounts of data, ranging from scientific measurements and life support data. These systems are quite complex and are dependent on the specific chosen architecture. Furthermore, there is no change in mass or power requirement over the period of time.

4. Thermal System Model

The thermal system of the lunar base is responsible for keeping the equipment and habitat at an ambient operating temperature to prevent overheating and malfunction. The model consists of cold plates, radiators and heat pumps. Radiative cooling is the primary method of heat rejection on the moon due to its lack of atmosphere. The radiators are responsible for dissipating heat produced within a base. Cold plates act as heat sinks that absorb thermal energy to help maintain desired operating temperature ranges for all components. Heat pumps are used to maximize the energy efficiency of the base while decreasing the dependence on radiators for excessive heat rejection [27, 28].

The equations to calculate the power draw and mass requirements for each of these components have been detailed in the following subsections. The heat transfer equation that is used across all components is as follows [29].

$$\begin{aligned} \text{Heat Load} &= \sigma \cdot \epsilon \cdot (T^4_{\text{Source}} - T^4_{\text{Sink}}), \text{ where} \\ \text{Heat Load} &= \text{Power Requirements of the System} \\ \sigma &= \text{Stefan-Boltzmann's Constant} = 5.67 \times 10^{-8}, \\ \epsilon &= \text{Emissivity of the component under consideration,} \\ T_{\text{Source}} &= \text{Temperature of the heat source, and} \\ T_{\text{Sink}} &= \text{Temperature of the heat sink.} \end{aligned} \tag{3}$$

The heat loads, T_{Source}, and T_{Sink} are different for each component and thus this equation is used differently for each. It is assumed that electrical energy to thermal energy conversion is 100%. This allows for direct modeling of heat load from the system's power requirements.

Cold Plates

An average temperature for all components of all systems is considered. The T_{Source} for the cold plates is found by manipulating the governing equation. T_{Sink} is the ambient temperature maintained in the habitat. The heat load is the total nominal power requirement of the habitat components at any given time. T_{Source} found here is the average temperature of the machine during use. The specific heat load for the cold plates is found by using the average temperature of the component (set as source) and the desired operating temperature of the component, which may be higher than the ambient habitat temperature.

Once the heat load is known, the equation below can be used to calculate the mass of the cold plates [30]:

$$\text{Mass of Cold Plates} = \frac{43.54 \times \text{Calculated Heat Load}}{\text{Fluid Flow Rate} \times 5.552} \tag{4}$$

Cold plates act as a passive component in terms of the power consumption since they do not draw any power directly from the power system. As such, there is no separate power calculation for cold plates.

Heat Pumps

Heat pumps are active components of the thermal system. The power requirement of heat pumps is calculated using the following formulae [27]:

$$\text{Power Consumption of Heat Pumps} = \frac{\text{Fluid Flow Rate} \times \text{Pump Head}}{\text{Pump Efficiency} \times 550 \times 3600} \tag{5}$$

The mass of the heat pump is calculated based on the fluid flow rate in the pipes. A redundancy of 2 is applied to account for failure of one line.

$$\text{Mass of a Heat Pump} = 5.61 \times \left(\frac{\text{Fluid Flow Rate}}{\text{Fluid Density} \times 60}\right)^{\frac{3}{4}} \tag{6}$$

The value obtained from the above equations is multiplied by two to account for the redundancy and that gives us the final mass value.

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Radiators

The radiators do not draw any power from the power system and are passive components. The mass of radiators depends on the amount of heat that needs to be rejected across all systems and the operating temperature that the radiator is designed to. The amount of heat rejected is calculated using the heat transfer equation, wherein T_{Source} is the operating temperature of the radiator and T_{Sink} is the temperature of space that the radiators will be pointed towards; this value is 3K.

The rejected heat load is used to size the radiators using the following equation:

$$\text{Radiator Area} = \frac{\text{Heat Load of Cold Plates}}{\text{Heat Rejected by Radiator}} \tag{7}$$

The heat load on the cold plates indicates how much heat needs to be removed from the system. To effectively transfer this heat, radiator surface area needs to be appropriately sized in order to allow efficient heat dissipation.

$$\text{Mass of Radiator} = \text{Radiator Area} \times \text{Unit Area Radiator Panel Mass} \tag{8}$$

The mass of a unit radiator panel is technology specific.

5. *Power System Model*

A Power System on the lunar base can be broken down into 3 subsystems, namely, power generation, power management and distribution (PMAD), and power storage.

There are several technologies that have been proposed for each subsystem. Z. Khan et.al [31] suggest several technology alternatives for early development of the base. Their study assumes the base is near Shackleton crater on the south pole, which is in agreement with NASA's determined location of the lunar base [32, 33]. The presented technology alternatives were evaluated against the attributes of constructability, maintainability, reliability, adaptability, system safety, technology gaps/risk and the alternatives are as follows[34]:

Table 2 Technology options for power generation, PMAD and power storage

Power Generation	Power Management & Distribution (PMAD)	Power Storage
Solar Arrays	Regenerative Fuel Cells	AC Transmission Lines
Nucelar Power	Batteries	DC Transmission Lines
	Flywheels	Optical Bending
	Thermal Reservoirs	Radio Frequency Power Beaming
		Light Bender

Based on the conclusions derived in [35], this study incorporates solar panel arrays for power generation, DC transmission lines for power distribution, and Li-Ion batteries for power storage.

Power Generation Model

The extreme conditions on the Moon, with their long periods of darkness and extreme temperatures, make it impossible to use traditional power sources. Solar panels guarantee uninterrupted electricity for life and scientific research on the lunar surface [31], and their high TRL for space applications make it a reliable technology to implement. The total power that needs to be generated depends on the total power draw from all the systems of the lunar base. The outputs of power requirements from all other systems act as an input for the power generation function. The total power is summed, which allows us to calculate the total area of solar panels that are required.

$$\text{Area of Photovoltaic Arrays} = \frac{\text{Power Required}}{\text{Generation Capability}} \tag{9}$$

The generation capability is given in W/m^2 are decided by the kind of solar panel arrays being used. The value used for solar irradiation for solar panels is 1372 W/m^2 [36]. The efficiency for these solar panels varies between 29 to 35% [37][38]. The mass of the solar panels is then found as a function of the area of solar panels required:

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$$\text{Mass of Arrays} = \text{Area of Panels} \times \text{Mass per m}^2$$

The unit mass of a solar panel is taken to be 2.25 kg/m^2 and is used to calculate the number of solar panels required.

Power Distribution and Management Model

The PMAD system for the lunar base relies on direct current (DC)transmission lines. The DC lines have a good transmission efficiency which is important for lunar operations. Specially designed DC transmission lines can handle voltages required by the lunar base's equipment. Robertson et al. [34] studied the effectiveness of different distribution systems on the lunar surface and concluded that DC lines are most effective when the distribution is in less than a 4 km range. The values for DC cable mass used in the framework were obtained from their study.

Power Storage Model

An advanced battery system ensures that energy is always available in the harsh lunar environment, especially when solar panels cannot generate power during the lunar night. These batteries are designed to handle extreme temperatures and radiation, and this study incorporates lithium-ion (Li-Ion) batteries. They are assumed to be charged before launch, which reduces the power draw from the generation system. The power requirements of the systems during the lunar night is the power storage requirement. These values come from power outputs of other systems. The mass of batteries required to store that energy is then calculated using the total power to be provided, power rating per m^2, and battery efficiency [39].

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III. Sample Problem

The sample problem is structured to showcase the framework's capabilities, focusing on reporting output parameters essential for practical applications. Utilizing mass as a key criterion tied to mission costs, the framework closely monitors resource outputs, power needs and mass requirements across the mission duration. Power and mass are interdependent in that increased power requirements cause a substantial increase in the mass of the power system. For an architecture, the power requirements are affected the most due to the different power demands from lunar day to lunar night. Due to lunar night's constraints, reliable energy sources must be identified; battery solutions and alternative power production sites are key choices to this trade study. This sample problem uses an illumination map reader developed based on data about illumination conditions on the lunar south pole, to help select an optimal power architecture while ensuring consistent functioning of the remaining systems throughout the trade study setup. However, since the power generation depends on the illumination cycle, a favorable location for the generation site must be determined such that it is illuminated for a majority of the times. The process used to determine this location as well as the set up of the sample problem is described further in this section.

A. Illumination Map Reader to Shortlist Locations

NASA's illumination map data contains information about the day and night conditions in a 2° radius around the lunar south pole. The illumination map reader processes this data to find out the percentage time illumination of different locations. This helps determine the feasibility and requirements of setting up single or multiple power generation sites. The method used to read the illumination map is shown in Fig. 4. Figure 3 showcases a sample frame of the illumination data that has been used

The chosen period for our case study is January 1, 2025 - January 30, 2025. The method described in Fig. 4 is used to find the percentage of time a pixel is illuminated over this period. A heat map is then produced as a translation of this data into a visual format. This heat map is used to identify potential locations for the lunar base such that the base is in a highly illuminated region, but also close to a permanently shadowed region for scientific purposes [7] [40]. Additionally, for multiple power generation sites, a highly illuminated region should be within 4 km of another such that the light and dark periods of each site complement each other. Keeping the distance between the two sites within 4 km is necessary to maintain minimum mass of DC distribution lines[34].

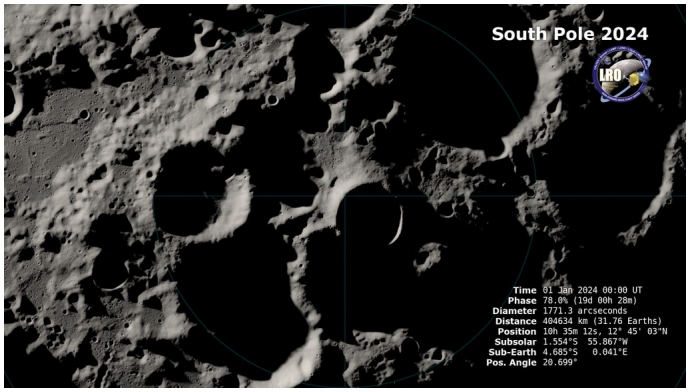


Fig. 3 Frame representing illumination conditions for 2° radius around the lunar south pole [9]

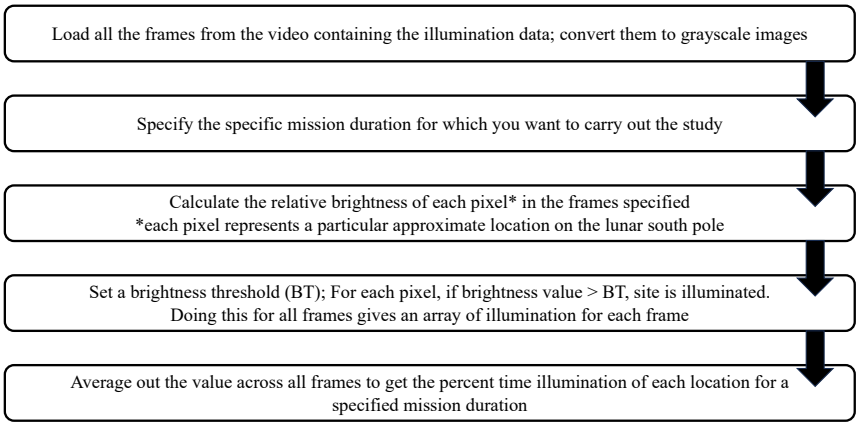


Fig. 4 Python modeling strategy to develop the illumination map reader

The heatmap in Fig. 5 shows several potential sites for the base along the rim of Shackleton Crater. To fulfil the requirements for a secondary power generation site, a time vs. illumination plot is generated for each of the high illumination regions, which are superimposed to see highest coverage sites.

The site highlighted in Fig. 6 is illuminated for 93% of the time and will be the site of one of the power generation sites. The framework identifies day and night based on the illumination patterns specific to this site. The time resolution for this illumination map is 2 hours and it is assumed that the illumination over the two hours is constant. The complementary site found for this location is as seen in Fig. 6.

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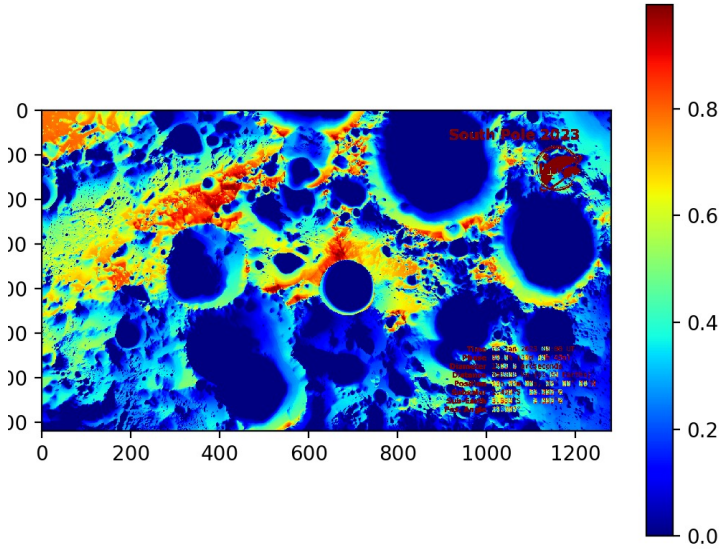


Fig. 5 Heat map representing the illumination duration (in percent time)

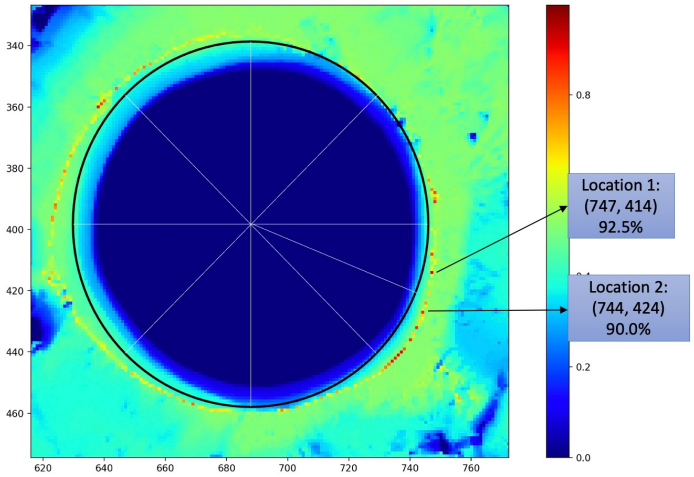


Fig. 6 Shortlisted locations from the illumination map reader for the placement of solar panel arrays

To ensure that the complementary site is within a 4 km range from the primary, the crater is divided into arcs of equal length to find distances along the crater rim. The diameter of Shackleton Crater is 19.3 km [41] and is used in the calculation as shown in Fig. 7

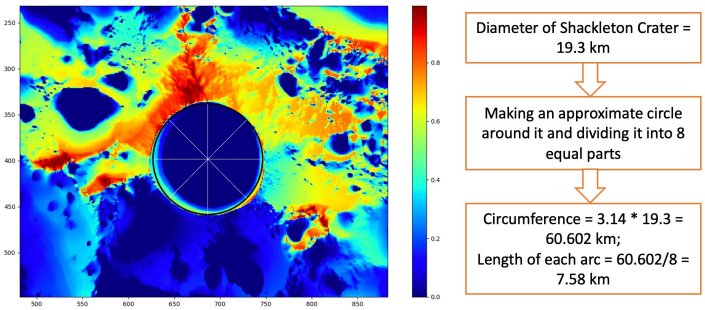


Fig. 7 Calculations to find out distance along the rim of Shackleton Crater

Analysis on the sites shown in Fig. 6 yields a distance of less than 4 km and 98.33% illumination. Thus, these two sites are used in the case study.

B. Sample Problem Setup

The sample problem focuses on the subsystem-level trade of having a singular power generation site with full reliance on battery usage during lunar night versus two complementary power generation sites with minimal usage of stored battery power. The power demands for lunar day and night differ to ensure that the variable power demands of the base over daytime and nighttime are captured, however, to isolate the mass of the two alternatives, all other parameters are kept constant.

The simulated mission consists of a crew of 4 on the lunar base for 30 Earth days. Nominal ECLSS functioning is considered. Within ECLSS, the Oxygen Generation Assembly (OGA) and Water Recovery System (WRS) run on cyclic operation modes. It is assumed that only fractions of products produced in a day are consumed in various activities, leaving ample stored product. For ISRU, it is assumed that the excavation activities will be carried out every day during the lunar day and every other day during lunar night to conserve power. This is not necessary for the operation of the base, but helps relax some constraints on the power generation and leaves room for future trade studies to be carried out on ISRU operation. Two regolith extraction rovers will be used to their full capacity during the day and to $\approx$ 70% capacity during the night time. Thus, the power generation is not sized to the maximum possible constraints, but gives a decent idea about mass values. A fraction of the water produced using the regolith is further used to produce oxygen. The thermal and power systems are sized to the requirements of all systems to accommodate loads during the day and night.

IV. Results

The results of the sample problem demonstrate both the functionality of the tool and help evaluate an important trade. The results focus on the power system mass as a consequence of lunar day and lunar night, with two configurations for power generation and storage.

Power required for base operation

The power system is sized to the highest constraints and is hence sized to lunar night power requirements. The total energy consumed by all systems during a lunar night (assuming longest duration of night in a 24 hour period) is 531.498 kWh. This translates to 22.15 kW of required power generation capabilities. Figure 8 shows the breakdown of the power requirement by system.

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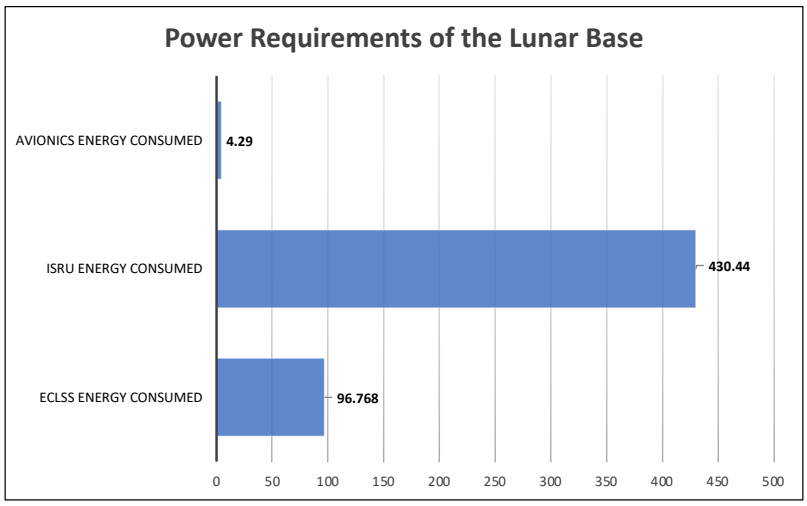


Fig. 8 Energy requirement breakdown of the lunar base

Singular power generation site

The single power generation site is located on the crater rim from the shortlisted sites shown in Figure 6. Fig. 9 shows the variation of the power generation capabilities of this power generation site as a function of time, which is a result of the illumination cycles at that site.

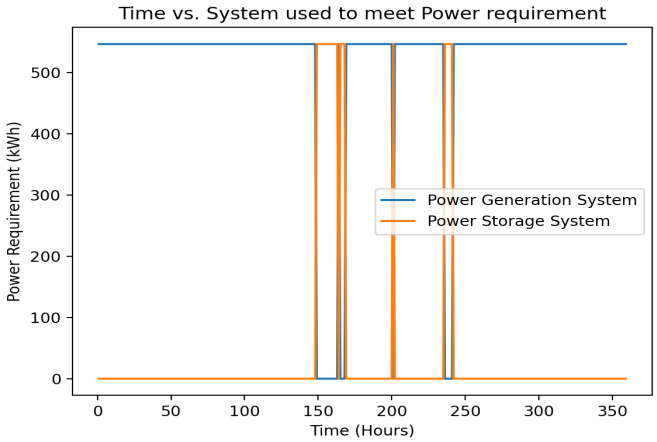


Fig. 9 Time vs. Power Requirements for the case of a singular power generation site

The blue line demonstrates generation by the solar panels when they are illuminated and a dip in the blue line indicates a shadow at that site. The spike in the orange line corresponds to the dip in the blue line indicating the usage of stored power. Accordingly, the mass of the battery storage depends on how long and how often it needs to supply power to the base since it entirely depends on the battery during lunar night.

Two complementary power generation sites

The two power generation sites in this case are both 4 km from the base and each other. These two sites together provide approximately 98.33% illumination coverage and the base relies on battery power for the remaining time.Fig. 10 shows the variation of the power generation capabilities of the base as a function of time.

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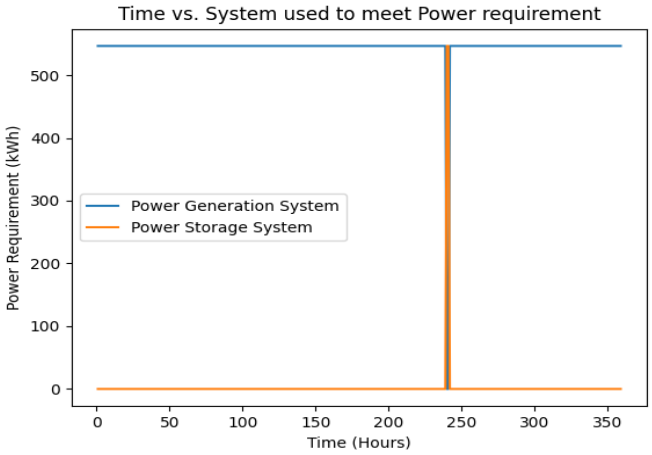


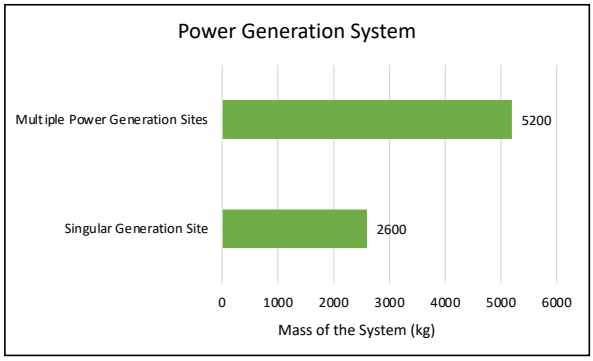
Fig. 10 Time vs. Power Requirements for the case of multiple power generation sites

Here as well, the blue line indicates that at least one of the two sites is producing enough power to power the entire base. The singular spike in the orange line indicates a period of approximately 6 hours when neither site is receiving light from the sun and therefore the stored battery power is being used.

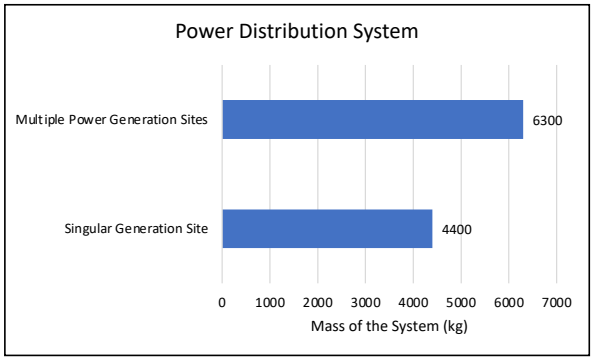
Mass Comparison

Plots shown in Fig. 9 and Fig. 10 indicate that multiple power generation sites result in minimal usage of the power storage system. The mass differs between the solar panels and PMAD system required for the second generation site, and the required battery mass sufficient for powering the base in the case of a singular generation site. Fig. 11 illustrates the difference between the two cases. Fig. 11 also shows a breakdown of the mass comparisons for each of the subsystems within the power system.

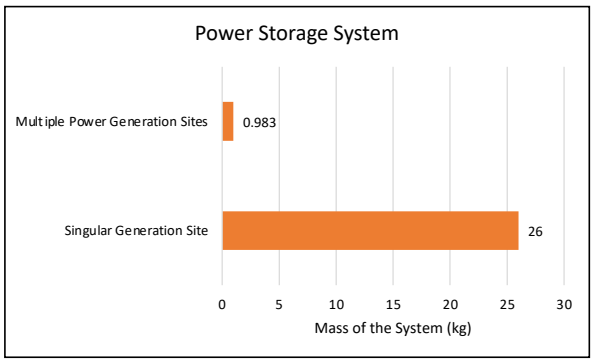
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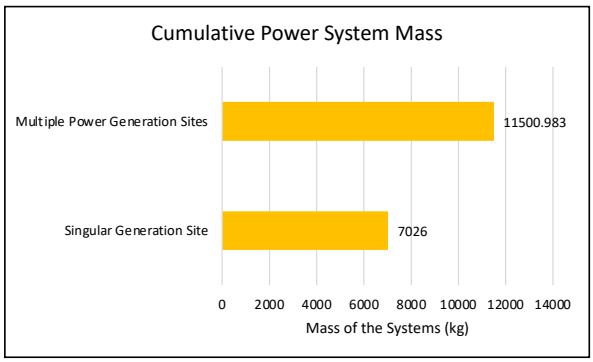
(a) Mass comparison of Power Generation Subsystem



(b) Mass comparison of PM&D Subsystem



(c) Mass comparison of Power Storage Subsystem



(d) Mass comparison of Cumulative Power System

Fig. 11 Mass comparison of the Power System

Fig. 11 shows negligible battery mass required for dual power generation sites, however, the overall mass of the power system is still higher than that of a single power generation site with high battery dependence. This is a result of significant increase in distribution lines through two generation sites, which are much heavier than the batteries. Based on the technology and data used for this study, it is more cost-effective to implement a singular power generation site with a power storage system in order to survive the lunar night. In a similar manner, many technology options can be compared against each other to find an ideal configuration that meets all requirements of a mission.

V. Conclusions

The system dynamics framework was effectively able to model the outputs from two power system trades as a function of time. The sample problem helped visualise time-dependency by demonstrating how frequently one would need to switch between the power generation and power storage systems. The framework also helped establish a mass comparison between the two cases; the significant mass difference points to the benefit of using a power storage system during lunar night, instead of implementing a secondary power generation system.

The base architecture in the sample problem, pertains to a specific configuration with specific technologies. It is also specific to the dates in consideration. A wide swath of illumination maps spanning a few years should be used for comprehensive analysis of appropriate sites and mass tradeoffs. Comparative studies for different configurations can be carried out by replacing technology parameters for technology options under consideration. The framework can help make comparisons between systems and within systems, their impacts on resource flows, and their corresponding costs. For example, a useful study could be to evaluate the necessity of employing both ISRU and ECLSS for life-support

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resource generation, and their corresponding contributions to those resources if both systems are employed in tandem. Several unique cases can be run without modifying any part of the framework itself.

The framework can be improved in multiple ways for future studies. The models should be modified to be able to take in arrays for individual parameters so that "real-time" dynamic behavior can be modeled. This allows for behaviors of subsystems to be variable during the mission period rather than have a singular operation mode for the entire duration of the mission. It would be beneficial to add the ability to assign locations for each of the systems or subsystems involved so that a detailed illumination-based study can be conducted. This is especially useful when considering ISRU plants that may not be directly next to the habitat and may therefore experience different illumination conditions.

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